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REG NO: 192511369

OUTPUT:

1)

```
MySQL 8.0 Command Line Client
mysql> CREATE TABLE STUDENT2
-> (
->   RegNo INT,
->   SName VARCHAR(20),
->   Gender CHAR(1),
->   DOB DATE,
->   MobileNo BIGINT,
->   City VARCHAR(15)
-> );
ERROR 1046 (3D000): No database selected
mysql> use dbms;
Database changed
mysql> CREATE TABLE STUDENT2
-> (
->   RegNo INT,
->   SName VARCHAR(20),
->   Gender CHAR(1),
->   DOB DATE,
->   MobileNo BIGINT,
->   City VARCHAR(15)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> ALTER TABLE STUDENT2
-> ADD PRIMARY KEY (RegNo);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student2;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| SName | varchar(20) | YES |     | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

2)

```
MySQL 8.0 Command Line Client
mysql> ALTER TABLE FACULTY
-> ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | NO | PRI | NULL    |       |
| FacName | varchar(15) | YES |     | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date | YES |     | NULL    |       |
| DOJ   | date | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| Dept  | varchar(10) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
```

```
MySQL 8.0 Command Line Client

mysql> ALTER TABLE FACULTY
-> ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | NO | PRI | NULL | |
| FacName | varchar(15) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| Dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> ALTER TABLE FACULTY
-> ADD CONSTRAINT chk_gender
-> CHECK (Gender IN ('M','F'));
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | NO | PRI | NULL | |
| FacName | varchar(15) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| Dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
```

3)

```
mysql> ALTER TABLE DEPARTMENT
-> DROP PRIMARY KEY;
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE DEPARTMENT
-> ADD PRIMARY KEY (DeptNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo | varchar(4) | NO | PRI | NULL | |
| DeptName | varchar(15) | YES | | NULL | |
| DeptHead | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

4)

```
mysql> alter table course
-> drop primary key;
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> ALTER TABLE COURSE
-> ADD PRIMARY KEY (CourseNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | PRI | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

```
MySQL 8.0 Command Line Client
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> ALTER TABLE COURSE
-> ADD CONSTRAINT chk_sem
-> CHECK (SemNo IN ('1','2','3','4','5','6'));
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | PRI | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

5)

```
6 rows in set (0.00 sec)

mysql> ALTER TABLE STUDENT2
-> ADD CONSTRAINT uq_sname
-> UNIQUE (sname);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student2
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| SName | varchar(20) | YES | UNI | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

6)

```
MySQL 8.0 Command Line Client

mysql> CREATE TABLE STUDENT3
-> (
->   RegNo INT,
->   Dept VARCHAR(15)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> ALTER TABLE STUDENT2
-> ADD PRIMARY KEY (RegNo);
ERROR 1068 (42000): Multiple primary key defined

mysql> ALTER TABLE STUDENT3
-> ADD CONSTRAINT FK_REGNO
-> FOREIGN KEY (RegNo)
-> REFERENCES STUDENT2(RegNo);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC STUDENT2;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| SName | varchar(20) | YES | UNI | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> DESC STUDENT3;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | YES  | MUL | NULL    |       |
| Dept  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql>
```

```
MySQL 8.0 Command Line Client
-> FOREIGN KEY (RegNo)
-> REFERENCES STUDENT2(RegNo);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC STUDENT2;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| SName | varchar(20) | YES | UNI | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> DESC STUDENT3;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | YES  | MUL | NULL    |       |
| Dept  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SHOW CREATE TABLE STUDENT3;
+-----+-----+
| Table | Create Table |
+-----+-----+
| STUDENT3 | CREATE TABLE `student3` (
  `RegNo` int DEFAULT NULL,
  `Dept` varchar(15) DEFAULT NULL,
  KEY `FK_REGNO` (`RegNo`),
  CONSTRAINT `FK_REGNO` FOREIGN KEY (`RegNo`) REFERENCES `student2` (`RegNo`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.00 sec)

mysql>
```